

SI Session 1 Review

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1 Solutions

1.1 Problem 1

$$\frac{1}{1+y^2} dy = \frac{x^3}{x^4+1} dx$$

$$\int \frac{1}{1+y^2} dy = \int \frac{x^3}{x^4+1} dx$$

$$\tan^{-1}(y) = \frac{1}{4} \ln(x^4+1) + K_1$$

$$\tan(\ln((x^4+1)^{1/4}) + K_1) = y$$

1.2 Problem 2

$$\frac{dy}{dx} = \frac{x}{x^2+1} \frac{1}{y \ln(y)}$$

$$\int y \ln(y) dy = \int \frac{x}{x^2+1} dx$$

Split the integrals

$$\int y \ln(y) dy = \frac{y^2 \ln(y)}{2} - \int \frac{y^2}{2y} dy$$

$$\int y \ln(y) dy = \frac{y^2 \ln(y)}{2} - \frac{y^2}{4} + K_1$$

$$\int \frac{x}{x^2+1} dx = \ln((x^2+1)^{1/2}) + K_2$$

$$\frac{y^2 \ln(y)}{2} - \frac{y^2}{4} = \ln((x^2+1)^{1/2}) + K_3$$

You could keep simplifying it, but this isn't an algebra class, so stopping here for this review is fine.

1.3 Problem 3

For sake of time, I'm going to skip a few algebraic steps.

$$\frac{dv}{dx}x + v = \frac{2x^2v}{x^2(1-v^2)}$$

$$\frac{dv}{dx}x = \frac{2v - v(1-v^2)}{1-v^2}$$

$$\int \frac{1-v^2}{v+v^3} dv = \int \frac{1}{x} dx$$

Separate Integrals

$$\int \frac{1-v^2}{v+v^3} dv = \int \frac{A}{v} + \frac{Bv+C}{v^2+1} dv$$

$$A = 1, \quad B = -2 \quad C = 0$$

$$\int \frac{1}{v} - \frac{2v}{v^2+1} dv = \ln|v| - \ln|v^2+1| + K_1$$

$$\int \frac{1}{x} dx = \ln|x| + K_2$$

$$\exp[\ln|v| - \ln|v^2+1|] = \exp[\ln|x| + K_3]$$

$$\frac{v}{v^2+1} = xK_4$$

$$yx = xK_4(y^2 + x^2)$$

1.4 Problem 4

Again, for sake of time I will be skipping some algebraic steps.

$$\frac{dv}{dx}x + v = \frac{x^2 + x^2v^2}{x^2v}$$

Rearrange to get v and x by themselves on one side

$$\frac{dv}{dx}x = \frac{1}{v}$$

$$\int v dv = \int \frac{1}{x} dx$$

$$\frac{v^2}{2} = \ln|x| + K_3$$

$$v = y/x$$

$$y^2 = 2x^2 \ln|x| + x^2 K_4$$

1.5 Problem 5

Test for exactness

$$\frac{\partial M}{\partial y} = 2x + 2y + \frac{1}{x} + \frac{1}{y^2 + 1}$$

$$\frac{\partial N}{\partial x} = 2x + 2y + \frac{1}{x} + \frac{1}{y^2 + 1}$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\int M(x, y) dx = \int (2xy + y^2 + \frac{y}{x} + \tan^{-1}(y)) dx$$

$$F = x^2y + xy^2 + \ln|x|y + x \tan^{-1}(y) + h(y)$$

$$\frac{\partial F}{\partial y} = x^2 + 2xy + \ln(x) + \frac{x}{1+y^2} + h'(y)$$

Now solve for $h(y)$

$$h'(y) = 0$$

$$\int h'(y) dy = 0$$

$$h(y) = K_1$$

Plug into F

$$K_2 = x^2y + xy^2 + \ln|x|y + x \tan^{-1}(y)$$

1.6 Problem 6

$$xy' + y = x^4$$

$$x \frac{dy}{dx} + y = x^4$$

$$\frac{dy}{dx} + \frac{y}{x} = x^3$$

$$IF = e^{\int (1/x) dx} = x$$

$$\int d(xy) = \int x^4 dx$$

$$y = \frac{x^4}{5} + \frac{K_1}{x}$$

1.7 Problem 7

$$y' + \frac{2x}{1+x^2}y = \frac{e^x x}{1+x^2}$$

$$IF = e^{\int \frac{2x}{x^2+1} dx} = 1+x^2$$

$$\int d((1+x^2)y) = \int x e^x dx$$

$$(1+x^2)y = e^x x - e^x + K_1$$

$$y = \frac{e^x x - e^x + K_1}{(1+x^2)}$$